

CHUKWUEMEKA OSMUND AZODO

MACHINERY ENGINEER | ROTATING EQUIPMENT, CONTROLS & RELIABILITY | OIL & GAS

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B.Eng Mechanical Engineering | ASQ Certified Reliability Engineer (CRE) | COREN R.Engr | ASME, SPE, NSE, ASQ

22 September 2026

Recruitment Team
ExxonMobil Nigeria
Esso Exploration and Production Nigeria Limited, Lagos

Application — Machinery Engineer, Nigeria Deepwater Assets

Dear Hiring Team,

I am applying for the Machinery Engineer position supporting the Nigeria Deepwater assets. I am a mechanical engineer with over ten years in machinery and rotating equipment engineering, currently the Machinery Engineer for the Oso Gas Hub — the largest offshore gas processing facility in the Seplat joint venture, and part of the former Mobil Producing Nigeria asset base. The standards, work processes and much of the machinery I work to every day are ExxonMobil-legacy, so I would join already fluent in how your assets are engineered and governed.

Safeguarding machinery reliability and availability is the substance of my current role. I review equipment performance daily — vibration spectra and trends, lube-oil analysis, compressor performance maps, turbine parameters and pump curves across HP and LP centrifugal compressors, turbo-expanders, gas turbines, cryogenic and API 610 pumps — through SolarInsight, Seeq and PI XHQ, and I issue the analysis and the recommendation to operations and maintenance whenever a machine drifts from its expected envelope.

I act as technical lead for machinery troubleshooting and recovery. When a bundle change-out left a 130 MMSCFD compressor with severe vibration, I led the root cause analysis with the OEM and in-house teams, mapped the contamination paths and ran sectioned borescope inspection of the cooler bundles; the cause proved to be debris-induced rotor unbalance, and piping cleaning, lube-oil flushing and re-inspection returned the machine to service within ISO vibration limits. A separate fleet-wide RCFA — fifteen years of records, using pooled failure-rate and Crow-AMSAA reliability-growth analysis — let me show the affected compressors failing roughly 17× more often than a comparable benchmark unit on the same platform, and trace the cause to a wrong-specification filter element fitted into a housing built for now-discontinued OEM hardware. I am now driving the closeout: OEM and alternative-supplier engagement on a validated fix, packaged into an eleven-action plan with named owners and dates.

Alongside that, I review equipment and system changes, specifications and deviations; contribute to risk screening, HAZID and detailed risk assessments; witness machinery Factory Acceptance Tests; and support commissioning and start-up — I developed and now steward the Pre-Startup Safety Review process for turbines and compressors at Oso, and was lead mechanical and reliability engineer for the Imo River Associated Gas Gathering plant start-up. I also build the digital side of the work: IIoT condition monitoring that cut operating cost by around ten per cent.

My offshore experience to date is on fixed platforms rather than FPSOs. The machinery scope transfers directly — gas turbine driven compression, turbo-expanders, API 610 and cryogenic pumps, lube-oil and seal systems, and Honeywell CCC anti-surge and turbomachinery control — and I would come up the curve on the floating production context quickly, particularly given the shared engineering heritage of the assets.

I hold a B.Eng in Mechanical Engineering and the NYSC discharge certificate, am an ASQ Certified Reliability Engineer and a Category I vibration analyst, and carry a valid BOSIET with CA-EBS and an Offshore Safety Permit. I am based in Nigeria and available at one month's notice.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience can support machinery reliability and availability across the Nigeria Deepwater FPSOs.

Yours sincerely,

Chukwuemeka Osmund Azodo